INCH-POUND
MIL-DTL-24643/69
3 March 2008

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LSC264

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

REQUIREMENTS:

Qualification Required.

Construction (Watertight) (see figure 1)

First	-	Solid copper conductor, tinned, in accordance with ASTM B286, 0.068±0.001 inch nominal outside diameter (O.D.).
Second	-	Thermoset insulation, diameter 0.176±0.004 inch nominal O.D.
Third	-	Braided shield (outer conductor) AWG no. 36 bare copper. Constructed using 16 carriers, 9 ends, 10±1 pick/inch. Coverage 92.5 percent minimum.
Fourth	-	Shield insulation of two polyester tapes, outer tape sealed. An alternate shield insulation may consist of one polyester tape plus an extruded jacket of a suitable transparent material, minimum average thickness of the transparent material 0.003 inch. The standard identification code shall be applied by Method 2 on the inner tape.
Fifth	-	Four coaxial components shall be cabled together with a 6±1 inch, left-hand lay.
Sixth	-	A binder tape applied at the manufacturer's option.
Seventh	-	Braided shield (outer conductor) AWG no. 34 tin-coated copper. Constructed using 24 carriers, 9 ends, 10±1 pick/inch. Coverage 90 percent minimum.
Eighth	-	Moisture barrier tape applied with overlap.
Ninth	-	Cross-linked polyolefin jacket with a 0.750±0.015 inch outer diameter, jacket wall thickness 0.070 inch minimum.

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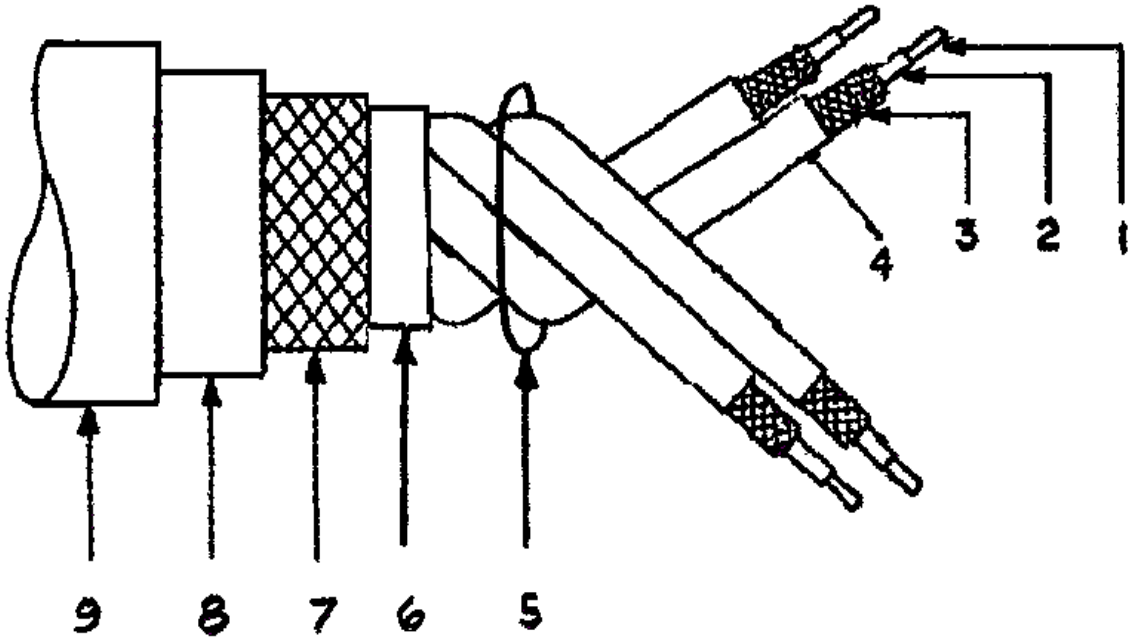
TABLE I. Details.

Military Part No. M24643/69	Type and Size	Overall Diameter (inches)	
		Min.	Max.
-01UO	LSC264-4	0.735	0.765



NOTE: Numbers on figure 1 refer to the construction steps.

FIGURE 1. Construction.

EXAMINATION AND TESTS:

Basic Electricals:

Conductor resistance (ohms/1000 feet at 25 °C, max.)	2.5
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	5000
Conductor to shield	2500
Shield to shield (component shields)	500
Shield to shield (component shield to overall (O/A) shield)	200
Insulation resistance (megohms/1000 feet, min.)	
Conductor to conductor	500
Conductor to shield	500
Shield to shield (component shields)	100
Shield to shield (component shield to O/A shield)	100

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Jacket flaws	No Failure
Conductor and shield continuity	No Failure
Group A:	
Visual and dimensional	No Failure
Characteristic impedance (ohms, max.)	40±2
(Time Domain Reflectometer (TDR) method in accordance with MIL-DTL-17)	
Radio frequency (RF) transmission loss (at 400 MHz, max.)	10dB/100ft
(Measure in accordance with MIL-DTL-17)	
Watertightness – see MIL-DTL-24643 for limits for water leakage	No Failure
Group B:	
Thermoset proof test (percent, max.)	
Overall jacket (200 °C)	50
Insulation	50
Cold bending, cable mandrel diameter 9 inches	No Damage
Tear (pounds per inch thickness, min.)	35
Physicals (unaged)	
Insulation	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1300
Elongation (percent, min.)	160
Group C:	
Physicals (aged)	
Insulation	
Air oven	
Tensile strength (percent of unaged, min.)	80
Elongation (percent of unaged, min.)	80
Jacket (cable)	
Air oven	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Shrinkage	No Failure

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Heat distortion (percent of unaged, max.)	30
Permanence of printing (jacket) (cycles, min.)	125
Group D:	
Flame propagation (cable)	No Failure
Qualification Inspection:	
Humidity conditioning, in accordance with EIA 364-31B Method IV, shall be performed prior to determining compliance to all other requirements in this specification. NOTE: Humidity conditioning is waived if the construction of the cable excludes use of water-swelling fillers.	No Failure

Qualification inspection shall include basic electricals, groups A, B, C, and D, plus the following:

Aging stability (cable) (125±5 °C)	No Failure
(Test in accordance with MIL-DTL-17)	
Dimensional stability (cable) (85±2 ºC)	No Failure
(Test in accordance with MIL-DTL-17)	
Abrasion resistance (overall jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket (component and overall)	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	
Jacket (component and overall)	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index, max.	
Jacket (component and overall)	25
Fillers	45
Insulation	45
Toxicity index, max.	
Jacket (component and overall)	5
Fillers	5
Insulation	1.5

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Durometer (overall jacket) – Type A (hardness, min.)	80
Weathering (overall jacket)	No Failure

UNIT ORDERING LENGTHS: All sizes 500 feet (nominal).

Custodians:
Army – MI
Navy – SH

Preparing Activity:
Navy – SH
(Project 6145-2008-004)

Review Activities:
Army – AR, AV, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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